

EMBEDDED SOFTWARE DEVELOPER

PRAVEEN M

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SUMMARY

- Result-oriented Embedded Developer highly skilled in designing and building Embedded codes to increase efficient of applications. Adept at identifying client requirements and ensuring specifications are met by routine checks for maximum client satisfaction. Proficient at debugging minor fixes.
- To continue a challenging and rewarding career that offers the opportunity to utilize knowledge as Embedded software Developer.

EDUCATION QUALIFICATION

- **B.E Electronics and Communication Engineering**, Completed, March 2022. SACS MAVMM Engineering College, Madurai. Marks 80%.
- **12th**, Completed, March 2018. MAVMM Hr.Sec.School, Madurai. State Board – Marks 65.5%.
- **10th**, Completed, March 2016. St Mary's Hr.Sec.School, Madurai. State Board – Marks 77.7%.

WORKING EXPERIENCE

Virtual Technology, Madurai.
Embedded Developer,
05-06 -2022 - 30-11-2022

The Petal Automations, Madurai.
Embedded Developer,
05-07-2022 – Present.

CERTIFICATION COURSES

- Embedded System for Arduino
- PCB Design
- Achieved “**HACKTHON**” from Wikitechy and Microsoft Research Community Group, Chennai in 2019.
- Participated Internship in **INTERNET OF THINGS at SACS MAVMM ENGINEERING COLLEGE.**

SKILLS

Decision Making

Problem Solving

Leadership

Self-Motivation

Multitasking

1.Pig Repellent device

Team: 2

Responsibility: Writing code in PIC and PCB Assembly.

Technology used: PIC Micro Controller, 555 Timer.

Project Overview: Designed and implemented an **automated pig repellent system** using a **PIC microcontroller** and **high-frequency speakers** for crop protection in agricultural fields. The device generates **periodic sound every 5 minutes** during **night hours (6 PM – 6 AM)** to deter wild animals without human intervention.

2. Nurse Call Station (NCS)

Team: 2

Responsibility: Writing Code on nRF52 for iBeacon.

Technology used: Bluetooth Low Energy (BLE), nRF52833.

Project Overview: Developed a **long-range wireless nurse calling system** using **Zigbee communication** capable of covering up to **1500 meters indoors** across hospital floors. Designed a **low-latency emergency alert network** with multiple call units and a **central nursing station dashboard** for real-time monitoring and response. Implemented **double-press beacon trigger logic** to avoid false alarms and ensure reliable nurse notification

3. Asset Tracking (Beacon)

Team: 2

Responsibility: Writing Code on nRF52.

Technology used: Bluetooth Low Energy (BLE), nRF52833, MPU6050.

Project Overview: Developed a **BLE-based asset tracking system** using an **nRF52 SoC** integrated with **MPU6050 motion sensors** for **real-time movement detection and monitoring**. Managed **end-to-end system design**, including **hardware selection**, **firmware development**, and **custom BLE communication protocols** for reliable data transmission. Implemented **sensor fusion algorithms** to accurately capture and interpret motion data, improving tracking accuracy and asset visibility.

LANGUAGES

- English
 - Tamil
- Full Professional Proficiency*

ENGINEERING / TECHNOLOGY

Languages:

- C
- C++
- Python
- Embedded C

Database:

- MYSQL

IDE:

- MPLAB IDE
- Arduino IDE
- Keil uVision,
- Embedded Segger IDE
- Visual Studio Code

Software:

- Proteous
- Advanced Design System(ADS)
- Cadence OrCAD / Allegro

4. STEM Project

Team: 3

Responsibility: Firmware in STC15 Series and Testing.

Technology used: One wire Communication Protocol.

Project Overview: Engineered a **steam control and monitoring system** using the **STC15F series microcontroller** with the **1-Wire communication protocol** for reliable data exchange within steam components. Designed and developed both **hardware and firmware**, enabling **precise control, parameter monitoring, and responsive system operation**. Implemented **1-Wire protocol stack** for efficient single-bus communication, ensuring accurate sensor interfacing and control logic.

5. IOT LAB KIT V 2 & 2.1

Team: 3

Role: Embedded Software Developer.

Responsibility: Writing code in Raspberry Pi.

Technology used: IOT, UART, SPI, RS232 to USB, USB to TTL, RFID, I2S, Bluetooth, MQTT and I2C communication Protocols.

Project Overview: It is for student learning purpose. Learn about how to interface I/O and sensors modules with IOT.

6. Electronic Shelf Label

Team: 1

Role: Embedded Software Developer.

Responsibility: Writing code in nRF52810.

Technology used: Bluetooth Low Energy (BLE), nRF52810, E-Paper Display.

Project Overview: Developed a low-power Electronic Shelf Label (ESL) using **nRF52810 BLE SoC** running **Zephyr RTOS** to wirelessly update product pricing on an **SSD1680-based e-paper display**. Implemented a custom **BLE GATT service** for real-time data transfer from a gateway and built a **Zephyr SPI display driver** for e-paper control. Optimized firmware for **RAM-constrained nRF52810**, achieving stable BLE + display operation with deep-sleep power management. Enabled multi-device scalability and reliable wireless price updates suitable for retail automation.

Controllers:

- Nodemcu
- ESP32
- Arduino UNO and MEGA
- Arduino UNO Q
- Raspberry pi
- PIC16F716
- nRF52 Series
- nRF91
- STC15

RF Technology:

- Bluetooth Low Energy
- Wi-Fi

7. IoT BLE Gateway

Team: 1

Role: Embedded Software Developer.

Responsibility: Writing code in Raspberry Pi, nRF52832.

Technology used: IOT, Ethernet, Wi-Fi, UART, Bluetooth Low Energy, MQTT and HTTP Protocols.

Project Overview: Designed and developed a **multi-protocol IoT BLE Gateway** using **Raspberry Pi Zero** as the main controller to bridge **BLE devices to cloud platforms** via **MQTT and HTTP**. Implemented **UART communication with nRF52-based modules** for controlled BLE scanning, beacon data acquisition, and ESL device management. The gateway supports **asset tracking, beacon monitoring, and Electronic Shelf Label (ESL) control**, similar to commercial gateways enabling centralized device management and real-time data upload to the cloud.

8. Web Based App Lab for Arduino UNO Q

Team: 4

Role: Embedded Software Developer (R&D).

Responsibility: Writing code in Python, UNO Q.

Technology used Wi-Fi, ADB Shell, SSH, AI.

Project Overview: Developed a **web-based Arduino development environment** with a **local edge agent** to enable secure **SSH access, ADB shell connectivity, and USB device control** from the browser. The local agent runs on the host machine, communicates with the cloud over the **local IP bridge**, and exposes hardware interfaces (Arduino UNO Q and other USB devices) to the web IDE. Implemented browser-to-device workflow for **code editing, compilation, deployment, and serial monitoring**, eliminating the need for a native desktop IDE.

ACADEMIC PROJECTS:

- ❖ We will create a micro strip patch antenna with specific characteristics, such as operating frequency, bandwidth, and gain. Once the antenna is built, we will conduct rigorous testing in an anechoic chamber to validate its real-world performance against simulated results. This will involve measuring radiation patterns, return loss, and other key parameters.
- ❖ This system utilizes Infrared (IR) technology to detect the presence of vehicles and control their engines automatically. In addition to the primary system functionality, we are incorporating a unique feature - the ability to generate driver resumes based on their waiting

PERSONAL INFORMATION

Father Name : Murugan.K
DOB : 26/12/1999
Gender : Male
Marital Status : Single
Nationality : Indian

DECLARATION

I do hereby declare that the above information is true to the best of my knowledge.

SIGNATURE
M.PRAVEEN